

CSE 472, Project I. Social Media Mining Content Related to the 2025 LA Wildfires

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1 Introduction

The Los Angeles Wildfires occurred in January of 2025 and devastated large parts of Eaton, the Palisades, and other L.A. suburbs, resulting in damages between \$76 billion and \$131 billion [3]. This project aims to use the social media platform *Mastodon* to gather and analyze data related to the disaster. Using *Mastodon*'s Python API library[1], two types of data were collected: posts, also known as "statuses" related to the disaster, and *Mastodon* users related in some way to the disasters. Collected statuses were used to create an information-diffusion network, characterizing the flow of information through the platform. Collected users were used to make a user network, demonstrating the connectedness of individuals related to the wildfires.

2 Data Collection

Mastodon's Python API was used to crawl posts and users related to the disaster. The sections below explain the methodology for each type of collection.

2.1 Keyword-Based Data Collection

To gather posts related to the L.A. wildfires, *Mastodon*'s hashtag timelines were searched for posts between Jan. 1, 2025 to Dec. 31, 2025 containing relevant keywords as tags. This time frame encompasses the actual disaster and ample time afterwards to capture information about the aftereffects of the fires. Table 1 shows the keywords used and the justification for each one. Most keywords used were some combination of an L.A. suburb followed by "fire(s)."

2.2 User-Based Data Collection

To gather users related to the L.A. wildfires, posts collected from keyword-based collection were used to help identify potential seed user candidates. A user was considered a potential seed user candidate if they had at least 1000 followers. Candidates were then investigated manually to narrow the selection further. Journalists, environmental scientists, and California residents were the primary users of interest. From there, the user list was expanded by viewing *@mentions* in collected posts belonging to each seed user. Since this method did not yield enough users, followers of seed users were added if they also had a relevant post from keyword-based collection. This was done to ensure no irrelevant users were accidentally collected. This process was done recursively to expand the seed user list, meaning the final list of collected users was a subset of the posters collected in keyword-based collection plus any extra users collected through *@mentions*.

Table 1: Keywords used in Keyword-Based Data Collection

Keyword	Justification
<i>#palisadesfire</i>	Name of Event
<i>#palisadesfires</i>	Name of Event
<i>#palisades</i>	Affected Area
<i>#Altadena</i>	Affected Area
<i>#Altadenafire</i>	Name of Event
<i>#Altadenafires</i>	Name of event
<i>#Eaton</i>	Affected Area
<i>#EatonFire</i>	Name of Event
<i>#EatonFires</i>	Name of Event
<i>#LAFires</i>	Name of Event
<i>#LAFire</i>	Name of Event
<i>#LAWildfires</i>	Name of Event
<i>#losangelesfires</i>	Name of Event
<i>#losangeleswildfires</i>	Name of Event
<i>#CaliforniaWildfires</i>	Name of Event
<i>#CaliforniaWildfire</i>	Name of Event
<i>#Californiafires</i>	Name of Event
<i>#Californiafire</i>	Name of Event
<i>#CAfires</i>	Name of Event
<i>#CAwildfires</i>	Name of Event
<i>#kennethfire</i>	Name of Event
<i>#sunsetfire</i>	Name of Event
<i>#lidiafire</i>	Name of Event
<i>#hurstfire</i>	Name of Event
<i>#hurstfire</i>	Name of Event
<i>#woodleyfire</i>	Name of Event

3 Network Construction and Visualization

3.1 Information Diffusion Network

The information diffusion network $I = (S, E_f)$ consists of the set of statuses S collected in keyword-based data collection. A directed edge $e_f \in E_f$ between two statuses (s_i, s_j) , $s_i, s_j \in S$ indicates the flow of information from s_i to s_j . This could indicate that s_j is either a repost of s_i or a reply to it. Note that, in the case of replies, the content of s_i is considered to elicit the response in s_j , hence why the direction of the edge is not reversed.

3.2 User Network

The user network $U = (P, E_u)$ consists of the set of users P collected in user-based data collection and the set of edges E_u representing interactions between users. We say that two users interact if at

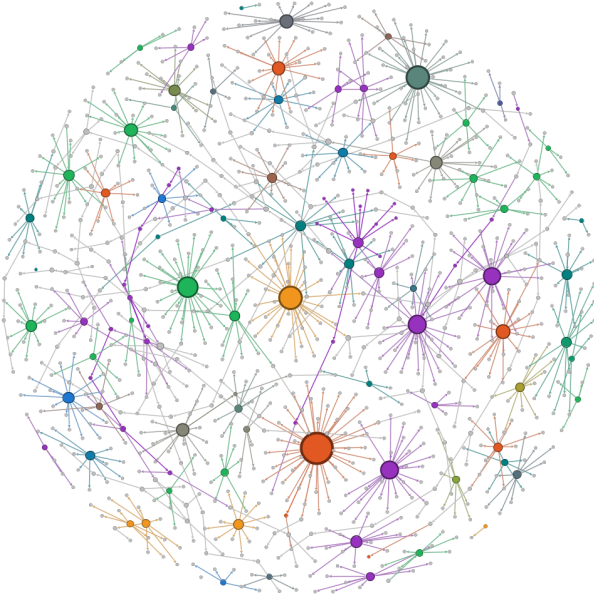


Figure 1: Information diffusion network containing posts, replies, and reblogs for keywords surrounding the 2025 LA Wildfires. Fructerman Reingold layout algorithm was used to achieve the dispersion of nodes. Nodes were scaled by out-degree. Coloring of the original posts is based on the combination of tags they contained. Colors for individual tag types are shown in Table 2.

Tag	Color
Contains <i>palisades</i>	Red
Contains <i>altadena</i>	Orange
Contains <i>eaton</i>	Blue
Contains <i>la</i> or <i>losangeles</i>	Purple
Contains <i>ca</i> or <i>california</i>	Green
Contains <i>fire</i>	Brown
Contains <i>other</i>	Grey

Table 2: Tag Colors

least one user replies to or reblogs the other user’s post. This representation is conservative since it does not take post likes into consideration, which would add more edges to the graph.

4 Network Measures

Five network measures were calculated for the user network: PageRank, EigenVector Centrality, Betweenness Centrality, average number of local relations, and average number of relations on the global level.

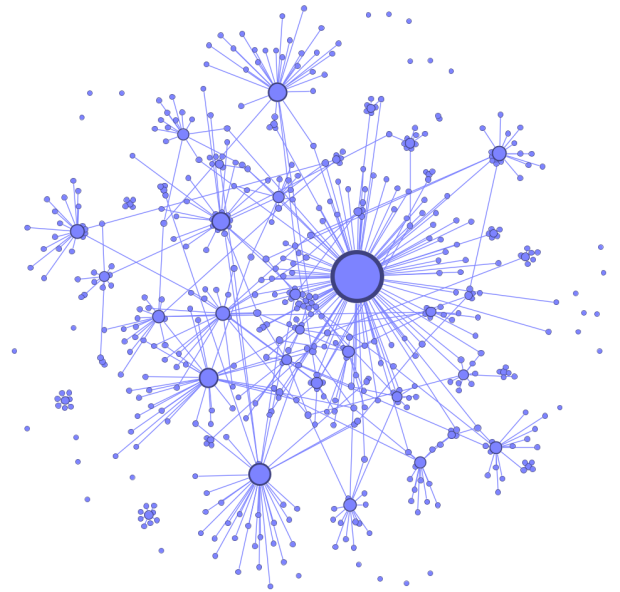


Figure 2: Graph representing user interactions regarding the LA Wildfires. Made with Gephi. A combination of Fructerman Reingold and Force-Atlas 2 (*LinLog Mode, Dissuade Hubs, and Prevent Overlap* set) layout algorithms were used to achieve the dispersions of nodes. Nodes were scaled by degree.

4.1 PageRank

The PageRank of a node v_i in an undirected graph is given by

$$PR(v_i) = \frac{1-d}{N} + d \sum_{j=1}^N \frac{PR(v_j)}{\deg(v_j)}$$

where N is the number of nodes in the graph, d is the dampening factor, which is typically set to 0.85, $PR(v_j)$ is the PageRank of a node v_j that points to v_i , and $d^+(v_j)$ is the out-degree of node v_j . PageRank is typically used to describe the importance of web pages, or how likely a user is to end up on a page after surfing for some time [2], but in this scenario it can be interpreted as a user’s structural importance in spreading information about the wildfires.

Figure 3 shows that the collective PageRank scores of the network resembles a power-law distribution, with most users having a very low score, and a small fraction of users having a much higher score—and very little in between. The extremeness of this distribution indicates that network interactions are highly concentrated, with a few central users reaching out to the rest of the population. However, it should be noted that this sample alone may not capture the actual network of interactions in its entirety.

4.2 Closeness Centrality

The closeness centrality of a node v_i is given by

$$C_C(v_i) = \left(\sum_{v_i \neq v_j} l_{ij} \right)^{-1}$$

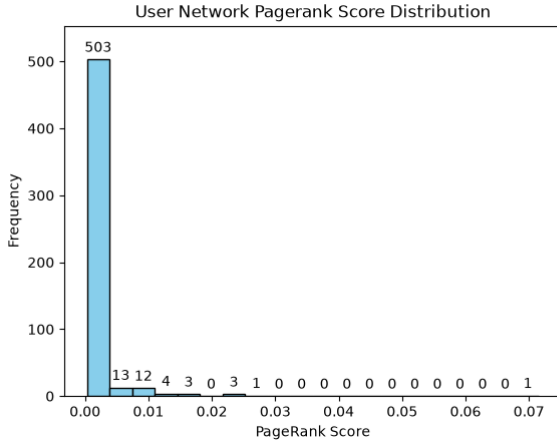


Figure 3: The PageRank score distribution of Mastodon users pertinent to the LA wildfires

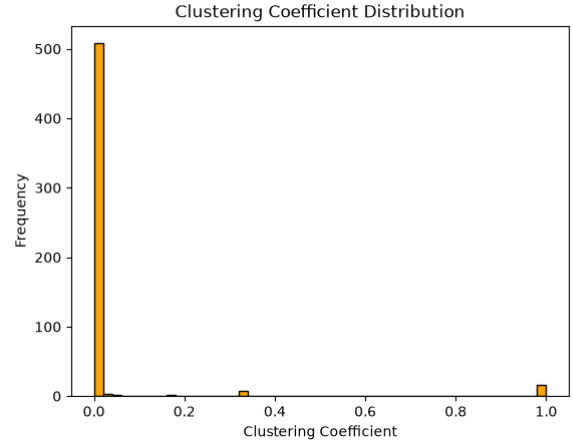


Figure 5: The clustering coefficient distribution of relevant Mastodon Users

4.3 Clustering Coefficient

The clustering coefficient of a node v is defined as

$$K(v) = \frac{\text{number of } v\text{'s neighbors that are connected}}{\binom{\deg(v)}{2}}$$

and represents how connected the neighborhood of a node is. Figure 4 shows that nearly all nodes have a very low clustering coefficient. The style of interactions, then, is more diffusive than circulatory. This suggests that users often interact with others who do not themselves interact with one another, rather than forming tightly interconnected groups. In the context of the wildfires, this might mean that information is being passed along without much back-and-forth conversation.

4.4 Average Relations

In this section we calculate both the local and global average number of relations. At a local level, for a node v_i , this is calculated as

$$R_l(v) = \frac{\deg(v_i) + \sum_{v_j \in N(v_i)} \deg(v_j)}{|N(v_i)| + 1}$$

This calculation considers the average number of relations only for a node and its immediate neighborhood.

Figure 4 shows that, while the majority of local average number of relations is less than 20, there exists a not insignificant subset of nodes that have a very high average number of local relations. This suggests that many "popular" Mastodon users, characterized by having many interactions, are separated from one another by at most two hops. Users with fewer interactions act mainly as connections to these popular users.

The global average number of relations in the user network, calculated as $\frac{2|E|}{|V|}$, is 2.13. This number alone would suggest that a user generally interacts with 2-3 other users. However, the local average distribution shows that this global average arises from a bimodal distribution rather than a normal one. In other words, the user-base is a combination of many sparsely connected users and a few users that are highly connected.

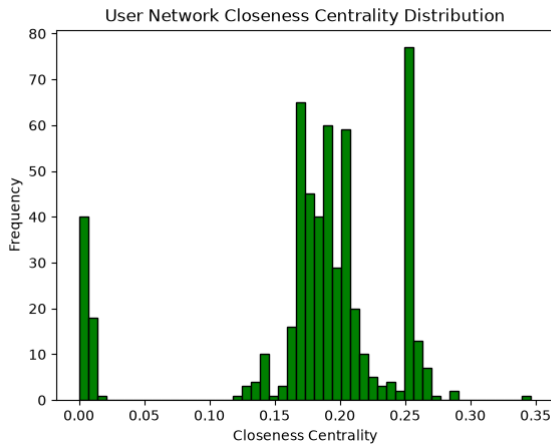


Figure 4: The distribution of closeness centrality in the user network

where l_{ij} is the length of the shortest path between v_i and v_j . In natural language, closeness centrality represents how quickly a node can reach every other node. In the user network, it shows how close a user is to other users through interaction chains.

Figure 4 shows a much more normal distribution of closeness centrality, albeit the mode is somewhat higher. The harmonic diameter of the graph is roughly 5.12 lengths, which may help provide a sense of scale for the centrality distribution. Overall, most users are only a few interaction hops away from any other user, meaning that information about the wildfires spread at a relatively fast pace.

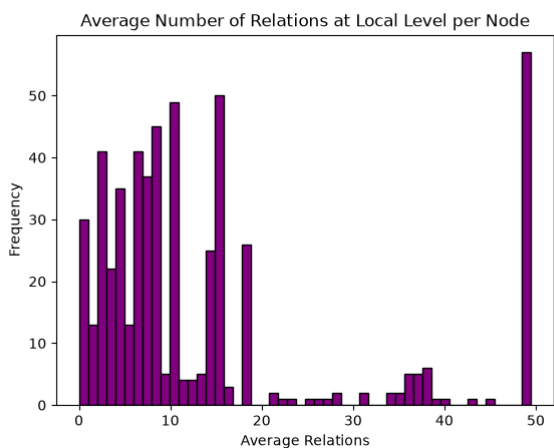


Figure 6: The distribution of the average number of relations on a local level per for each node.

5 Content Analysis

For collected Mastodon posts containing HTML content, unsloth/llama-3-8b-bnb-4bit was used to summarize the themes, impacts, and concepts of each post into keywords. Those keywords were then used to generate the word cloud shown in Figure 7. The exact instruction prompt used was:

"You will be given html content from a social media post about a natural disaster. Ouput exactly three keywords that summarize the content of the post. If possible, focus on identifying key disaster-related concepts, events, impacts, responses, and themes. Output the keywords as a comma separated list, and surround your response with the sequence '&&&'. For example:
&&&word.word.word&&&"

Among the collected keywords, stop-words defined by the word cloud library [4] were filtered out before generating the word cloud. Many of the keywords are directly related to the event, such as "fire" and "natural disaster," or mention a relevant location such as "California," or "los angeles." Among these directly related keywords, however, there are words which carry a stronger political sentiment, such as "climate change" and "Disinformation." Further research could investigate the link between political strife and the 2025 wildfires.

6 Conclusion

Two types of data surrounding the 2025 LA wildfires were collected from Mastodon and analyzed: posts and replies about the wildfires, and the corresponding account interactions. Analysis of user interactions suggests that most information about the disaster came from a select group of very popular users and propagated to many minor users in just a few hops. From the collected content, keywords were gathered using unsloth/llama-3-8b-bnb-4bit. Word cloud visualization highlighted that the themes of most posts

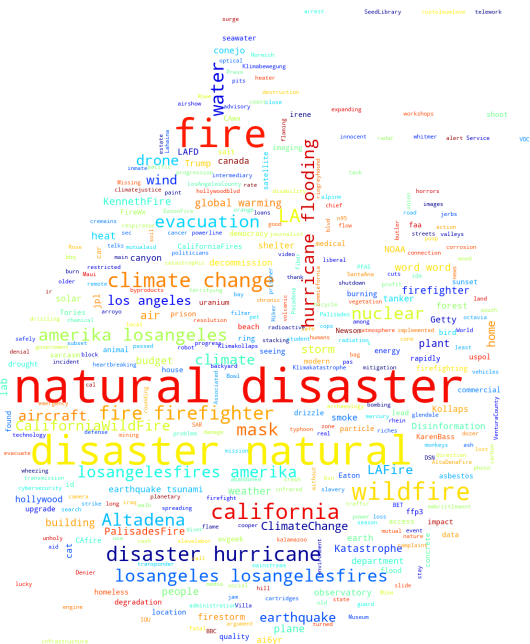


Figure 7: Word cloud summarizing the keywords in collected posts.

can be summarized by keywords directly describing the natural disaster and affected areas, and that there may be underlying political sentiment in posts about the wildfires.

References

- [1] Lorenz Diener and Corentin Delcourt. Mastodon.py: A python library for the mastodon API. 11(120):8946.
- [2] Cristian Leo. The algorithm that made google google.
- [3] UCLA Anderson School of Management. Economic Impact of the Los Angeles Wildfires, January 2025.
- [4] Andreas C Mueller. Wordcloud. original-date: 2012-11-04T22:57:59Z.

A Setting Up And Running The Code

See `Readme.md` for instructions on setting up and running the code used in this project.